

#setting home directory

```
getwd()
```

```
list.files()
```

```
setwd("C:/Users/USER/Downloads/R &  
Python")
```

##problem with source

```
.libPaths()
```

```
.libPaths("C:/Users/USER/Downloads/R &  
Python/pakage")
```

```
install.packages("ggplot2")
```

```
library(ggplot2)
```

Load dataset

```
flight <-  
read.csv("C:/Users/USER/Downloads/R &  
Python/flight_data.csv")
```

Random sample using student ID

```
set.seed(66)
```

```
flight_sample <- flight[sample(nrow(flight),  
10000), ]
```

##Question 1 :What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does \$500 seem like a good price for a coach ticket?"

Solution:

Summary statistics

```
summary(flight_sample$coach_price)
```

Output

```
Min. 1st Qu. Median Mean 3rd Qu.  
84.5 331.3 380.3 376.0 426.4  
Max.  
586.0
```

Mean

```
mean(flight_sample$coach_price)
```

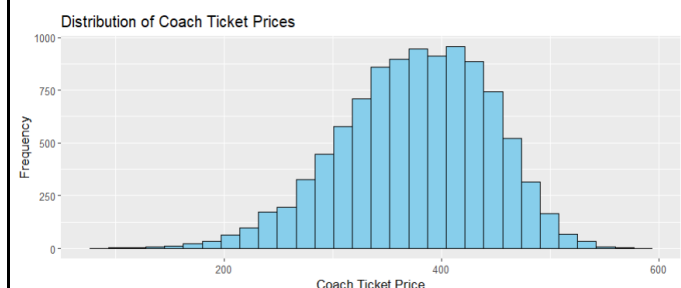
Output

```
[1] 375.989
```

Histogram

```
ggplot(flight_sample, aes(x=coach_price)) +  
  geom_histogram(bins=30, fill="skyblue",  
color="black") +  
  labs(  
    title="Distribution of Coach Ticket Prices",  
    x="Coach Ticket Price",  
    y="Frequency"  
  )
```

Output



Interpretation

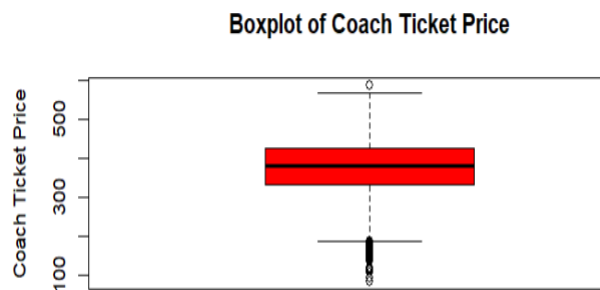
The histogram and summary statistics show the overall distribution of coach ticket prices. The average price can be compared with \$500 to determine whether it is considered expensive or reasonable.

#Boxplot

```
boxplot(flight_sample$coach_price,  
        main = "Boxplot of Coach Ticket Price",  
        ylab = "Coach Ticket Price",  
        col = "red",  
        border = "black")
```

Output

```
library(ggplot2) library(ggplot2)
```



##Q2: What are the coach prices for flights that are 8 hours long? Does a \$500 ticket seem more reasonable for these flights?

Solution:

#8-hour flights

```
flight_8 <- subset(flight_sample, hours == 8)
```

Summary statistics

```
summary(flight_8$coach_price)
```

Output

Min.	1st Qu.	Median	Mean	3rd Qu.
213.0	383.1	426.2	426.8	477.4
Max.				
568.5				

Interpretation :

Long-duration flights generally have higher ticket prices. Therefore, a \$500 coach ticket may be more acceptable for 8-hour flights.

##Q3: What does the distribution of flight delays look like? Are there any outliers?

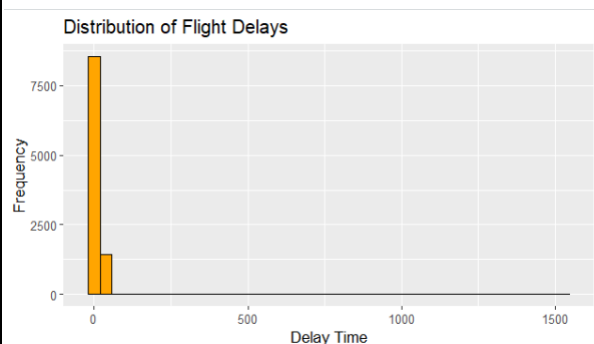
Solution:

```
library(ggplot2)
```

Histogram

```
ggplot(flight_sample, aes(x=delay)) +  
  geom_histogram(bins=40, fill="orange",  
                 color="black") +  
  labs(  
    title="Distribution of Flight Delays",  
    x="Delay Time",  
    y="Frequency"  
  )
```

Output



Summary

```
summary(flight_sample$delay)
```

Output

```
Min. 1st Qu. Median Mean 3rd Qu.
 0.00  9.00 10.00 13.29 13.00
Max.
1528.00
```

Interpretation :

The delay distribution may contain outliers due to extreme delays experienced by some flights.

##Q4: How are coach ticket prices related to flight distance (miles)?

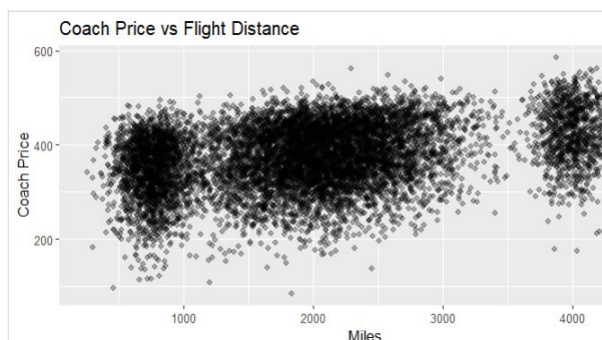
Solution:

```
library(ggplot2)
```

```
# Scatter plot
```

```
ggplot(flight_sample,
       aes(x=miles, y=coach_price)) +
  geom_point(alpha=0.3) +
  labs(
    title="Coach Price vs Flight Distance",
    x="Miles",
    y="Coach Price"
  )
```

Output



Correlation

```
cor(flight_sample$miles,
    flight_sample$coach_price)
```

Interpretation

There is usually a positive relationship between flight distance and ticket price.

##Q5: How are flight hours related to coach ticket prices?

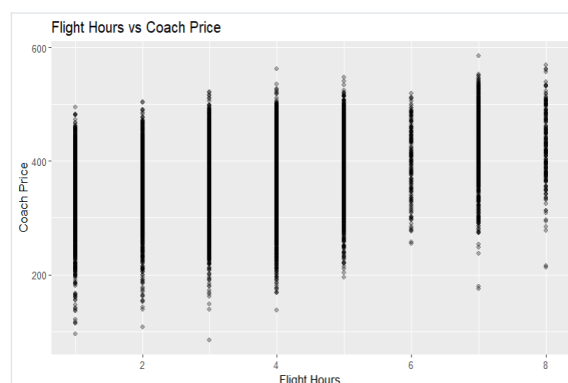
Solution:

```
library(ggplot2)
```

```
# Scatter plot
```

```
ggplot(flight_sample,
       aes(x=hours, y=coach_price)) +
  geom_point(alpha=0.3) +
  labs(
    title="Flight Hours vs Coach Price",
    x="Flight Hours",
    y="Coach Price"
  )
```

Output



Correlation

```
cor(flight_sample$hours,
    flight_sample$coach_price)
```

Output

```
[1] 0.3371431
```

Interpretation

Longer flight duration tends to increase coach ticket prices.

##Q6: How are coach ticket prices related to first-class ticket prices?

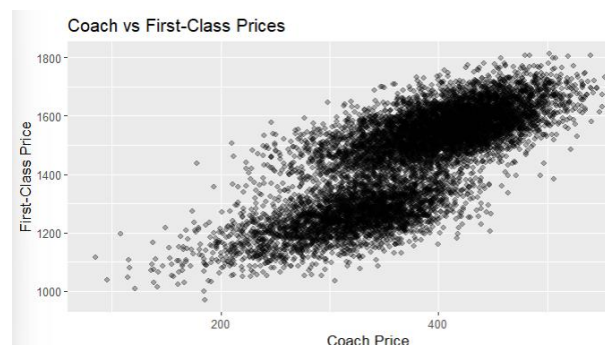
Solution:

```
library(ggplot2)
```

Scatter plot

```
ggplot(flight_sample,
       aes(x=coach_price,
           y=firstclass_price)) +
  geom_point(alpha=0.3) +
  labs(
    title="Coach vs First-Class Prices",
    x="Coach Price",
    y="First-Class Price"
  )
```

Output



Correlation

```
cor(flight_sample$coach_price,
    flight_sample$firstclass_price)
```

Output

```
<[1] 0.7625297 (strong positive) >
```

Interpretation

Higher coach prices are generally associated with higher first-class prices.

##Q7: How do in-flight features affect coach ticket prices?

Solution:

Meal

```
aggregate(coach_price ~ inflight_meal,
          data=flight_sample,
          mean)
```

Output

```
<inflight_meal coach_price>
1      No    369.4530
2     Yes    390.9813>
```

Entertainment

```
aggregate(coach_price ~
inflight_entertainment,
          data=flight_sample,
          mean)
```

Output

```
inflight_entertainment coach_price
1                      No    319.6572
2                      Yes    389.5651
```

Wifi

```
aggregate(coach_price ~ inflight_wifi,
          data=flight_sample,
          mean)
```

Output

```
inflight_wifi coach_price
1           No    311.4042
2           Yes    383.3489
```

Interpretation

Flights with additional services such as meals, entertainment, and WiFi often have higher ticket prices

##Q8: How do weekend flights compare to weekday flights in terms of ticket prices?

Solution:

```
library(ggplot2)
```

```
install.packages("dplyr")
```

```
library(dplyr)
```

Average prices

```
weekend_price <- flight_sample %>%
  group_by(weekend) %>%
  summarise(
    avg_coach = mean(coach_price),
    avg_firstclass = mean(firstclass_price)
  )
```

```
print(weekend_price)
```

Output

```
# A tibble: 2 × 3
```

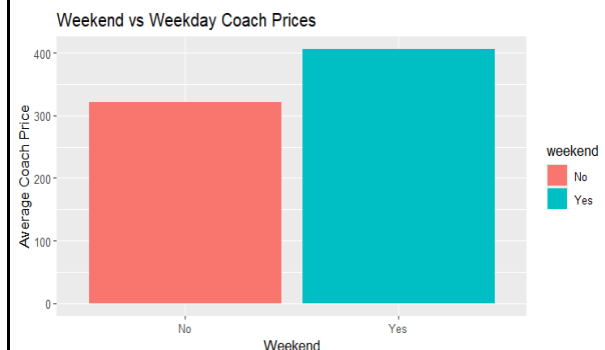
```
weekend avg_coach avg_firstclass
<chr>    <dbl>    <dbl>
```

```
1 No      321.    1259.
2 Yes     406.    1559.
```

Bar plot

```
ggplot(weekend_price,
       aes(x=weekend,
           y=avg_coach,
           fill=weekend)) +
  geom_bar(stat="identity") +
  labs(
    title="Weekend vs Weekday Coach Prices",
    x="Weekend",
    y="Average Coach Price"
  )
```

Output



Interpretation

Weekend ticket prices may differ from weekday prices depending on passenger demand

##Q9: How do redeye flights compare to non-redeye flights in terms of coach ticket prices?

Solution :

```
library(ggplot2)
```

```
library(dplyr)
```

```
# Average prices
```

```
redeye_price <- flight_sample %>%
```

```
  group_by(redeye) %>%
```

```
  summarise(
```

```
    avg_coach = mean(coach_price)
```

```
  )
```

```
print(redeye_price)
```

Output

```
# A tibble: 2 × 2
  redeye avg_coach
  <chr>   <dbl>
1 No      379.
2 Yes     318.
```

```
# Bar plot
```

```
ggplot(redeye_price,
```

```
  aes(x=redeye,
```

```
      y=avg_coach,
```

```
      fill=redeye)) +
```

```
geom_bar(stat="identity") +
```

```
labs(
```

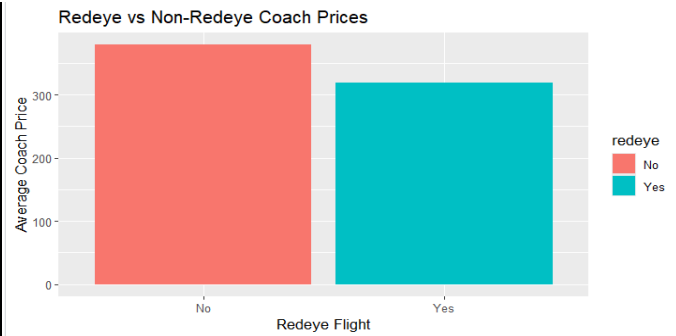
```
  title="Redeye vs Non-Redeye Coach
Prices",
```

```
  x="Redeye Flight",
```

```
  y="Average Coach Price"
```

```
)
```

Output



Interpretation

Redeye flights are often cheaper because they operate during late-night hours.

##Q10: Perform the analysis for appropriate variables

Solution :

```
#Summary statistics
```

```
summary(flight_sample)
```

Output

```
      miles      passengers      delay
Min.   : 260   Min. :141.0   Min. : 0.00
1st Qu.:1340   1st Qu.:204.0   1st Qu.: 9.00
Median :1978   Median :210.0   Median :10.00
Mean    :2001   Mean :207.6   Mean :13.29
3rd Qu.:2468   3rd Qu.:215.0   3rd Qu.:13.00
Max.    :4545   Max. :244.0   Max. :1528.00
inflight_meal  inflight_entertainment inflight_wifi
Length:10000   Length:10000   Length:10000
Class :character Class :character Class :character
Mode :character Mode :character Mode :character
```

```
day_of_week      redeye      weekend
Length:10000     Length:10000   Length:10000
Class :character Class :character Class :character
Mode :character Mode :character Mode :character
```

```
coach_price firstclass_price  hours
Min. : 84.5   Min. : 970.7   Min. :1.000
1st Qu.:331.3 1st Qu.:1300.6 1st Qu.:2.000
Median :380.3 Median :1501.2 Median :4.000
Mean :376.0   Mean :1453.7   Mean :3.622
```

3rd Qu.:426.4 3rd Qu.:1581.8 3rd Qu.:4.000
 Max. :586.0 Max. :1812.7 Max. :8.000

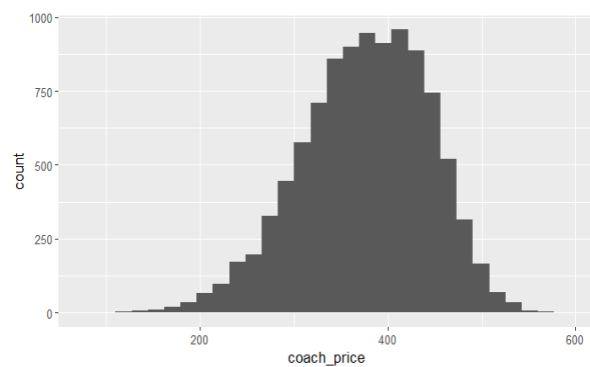
#(b) Visualization

```
library(ggplot2)
```

Histogram

```
ggplot(flight_sample,
       aes(x=coach_price)) +
  geom_histogram(bins=30)
```

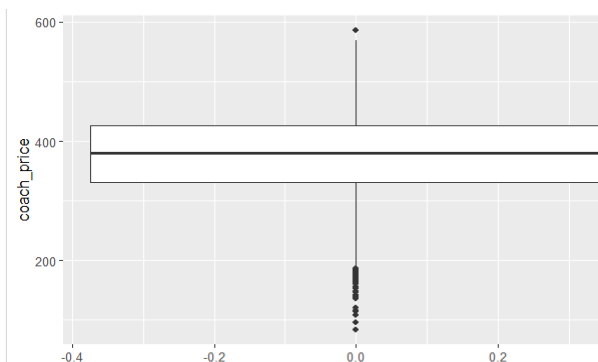
Output



Boxplot

```
ggplot(flight_sample,
       aes(y=coach_price)) +
  geom_boxplot()
```

Output

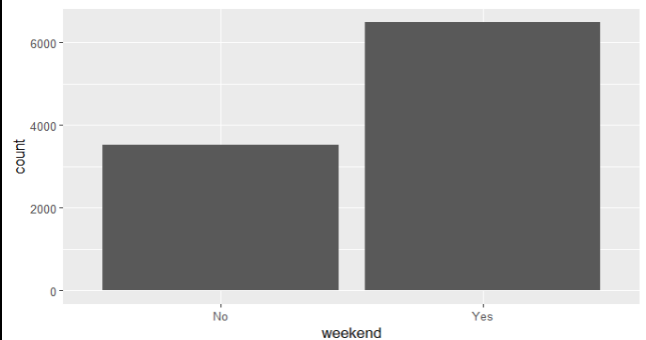


Bar chart

```
ggplot(flight_sample,
```

```
  aes(x=weekend)) +
  geom_bar()
```

Output



##(c) Hypothesis Testing (Independent t-test)

```
t.test(
  coach_price ~ weekend,
  data=flight_sample
)
```

Output

Welch Two Sample t-test

data: coach_price by weekend
 t = -73.684, df = 7064.8, p-value <

2.2e-16

alternative hypothesis: true difference in means between
 n group No and group Yes is not equal to 0

95 percent confidence interval:

-87.21771 -82.69724

sample estimates:

mean in group No mean in group Yes
 320.8771 405.8346

##(d) Correlation Analysis

```
cor(
  flight_sample[, c(
    "coach_price",
    "miles",
```

```
"hours",
"delay",
"firstclass_price"
)]
)
```

Output

	coach_price	miles
coach_price	1.0000000	0.342708985
miles	0.3427090	1.000000000
hours	0.3371431	0.985211774
delay	-0.0153009	-0.005755326
firstclass_price	0.7625297	0.295900806

	hours	delay
coach_price	0.33714307	-0.015300901
miles	0.98521177	-0.005755326
hours	1.00000000	-0.004757490
delay	-0.00475749	1.000000000
firstclass_price	0.29146142	-0.025528080

	firstclass_price
coach_price	0.76252966
miles	0.29590081
hours	0.29146142
delay	-0.02552808
firstclass_price	1.00000000

##(e) Linear Regression

```
model <- lm(
  coach_price ~ miles + hours + delay,
  data=flight_sample
)
```

```
summary(model)
```

Output

Call:
lm(formula = coach_price ~ miles + hours + delay, data = flight_sample)

Residuals:

Min	1Q	Median	3Q	Max
-287.736	-42.123	4.627	49.941	178.604

Coefficients:

	Estimate	Std. Error	t value
(Intercept)	326.852093	1.512646	216.080
miles	0.025869	0.003952	6.546
hours	-0.647266	2.145323	-0.302
delay	-0.021212	0.014969	-1.417

Pr(>|t|)

(Intercept)	< 2e-16 ***
miles	6.21e-11 ***
hours	0.763
delay	0.157

Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 63.95 on 9996 degrees of freedom
Multiple R-squared: 0.1176, Adjusted R-squared: 0.1174
F-statistic: 444.2 on 3 and 9996 DF, p-value: < 2.2e-16

#10(f) Logistic Regression

```
flight_sample$redeye_binary <-
  ifelse(
    flight_sample$redeye=="Yes",
    1,0
  )
```

```
log_model <- glm(
  redeye_binary ~ hours + coach_price,
  data=flight_sample,
  family="binomial"
)
```

```
summary(log_model)
```

Output

Call:


```
glm(formula = redeye_binary ~ hours + coach_price, family = "binomial",
     data = flight_sample)
```

Coefficients:

```
      Estimate Std. Error z value
(Intercept) 1.2025056 0.2130049  5.645
hours        0.2716457 0.0276520  9.824
coach_price -0.0144507 0.0006844 -21.114
      Pr(>|z|)
(Intercept) 1.65e-08 ***
hours        < 2e-16 ***
coach_price  < 2e-16 ***
```

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 4254.0 on 9999 degrees of freedom
Residual deviance: 3767.7 on 9997 degrees of freedom
AIC: 3773.7

Number of Fisher Scoring iterations: 6

Conclusion

This analysis explored several important aspects of airline ticket pricing and flight characteristics using R programming. The study identified relationships between ticket prices, distance, flight duration, delays, and additional flight services. Statistical analysis and visualization techniques helped provide meaningful insights into airline pricing patterns.

References

- RProgramming Documentation
- ggplot2 Package Documentation
- dplyr Package Documentation